

NOVATEL ANALYTICS

Customer Churn Analysis

Stakeholder Insights Briefing

Prepared for: **Executive & Retention Leadership Team**

Data Sources: **SQL Server -> Power Query -> Power BI Semantic Model**

July 2026

Agenda

What we'll cover in this briefing



01

The Business Problem

Why churn matters and what it's costing NovaTel today



02

Data Architecture & Governance

SQL Server integration, modeling, and security behind this report



03

Key Findings

Who is churning, why, and where the revenue risk sits



04

Self-Service Controls

What-if parameters, dynamic metrics, and drill-down features built for stakeholders



05

Recommendations & Roadmap

Prioritized next steps and what we propose to build next

The Business Problem

Churn is eroding revenue faster than it's being replaced

Why this matters now

- Nearly half of NovaTel's customer base has churned — well above the 13% internal target.
- Month-to-month contracts and Premium-tier customers churn at the highest rates, driving avoidable revenue loss every month.
- Early-lifecycle customers (0-6 months) retain the worst — the leak starts before loyalty ever forms.
- Warning signs are visible before churn happens: 60+ day login inactivity is the single strongest predictor.

47%

Overall Churn Rate

132.65K

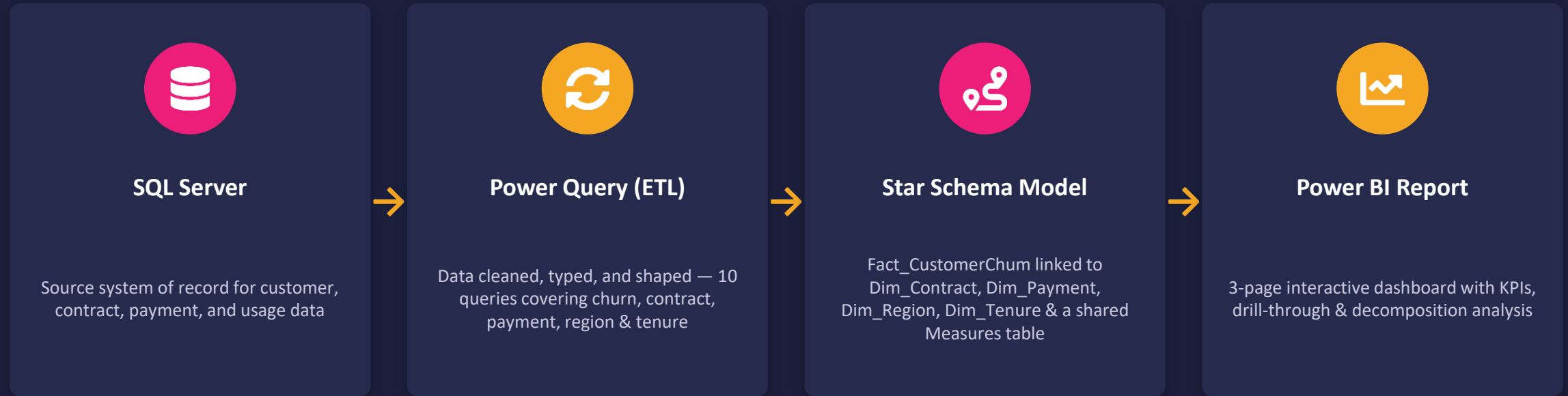
Monthly Revenue Lost

234 / 500

Customers Churned

How the Data Comes Together

From SQL Server to a governed, reusable analytics model



Why it matters: A single governed semantic model means every KPI — churn rate, retention rate, revenue at risk — is calculated once and reused consistently across every page, filter, and stakeholder view.

Built for Governance, Not Just Reporting

Modeling decisions that make this dashboard safe and scalable to hand to a real organization

Semantic Model Structure

Fact_CustomerChurn — Central fact table — one row per customer, incl. a pre-computed Risk_Score

Dim_Contract / Dim_Payment — Contract type, internet service, plan, and payment method attributes

Dim_Region / Dim_Tenure — Region hierarchy and tenure-band segmentation

User_region_Map — Maps each report user to their region — the basis for Row-Level Security

Measures_Table — Centralized DAX measures, reused consistently across all 3 report pages

Target Churn/Retention Rate % — What-If parameter tables driving the interactive target sliders



Row-Level Security Ready

The User_region_Map table lets this model restrict data by user — a Regional Manager would see only their own region, without duplicating the report.



Governed ETL in Power Query

Source data is cleaned and typed before it reaches the model — columns are standardized, conditional fields added, and irrelevant fields removed for a lean, trustworthy fact table.



Pre-Scored Risk Signal

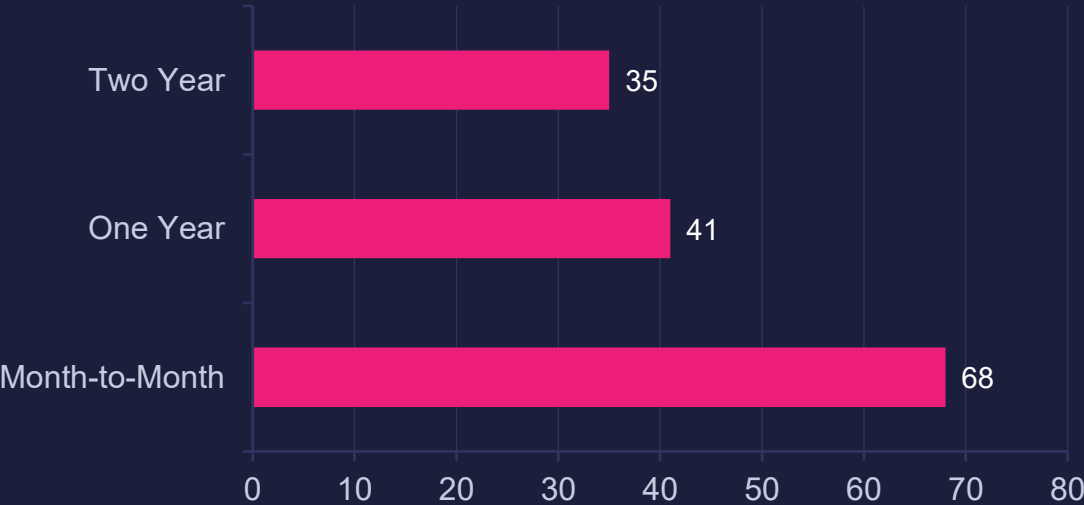
Risk_Score is calculated once upstream in the fact table, not recreated per visual — keeping every page's risk view consistent.

Executive KPI Snapshot

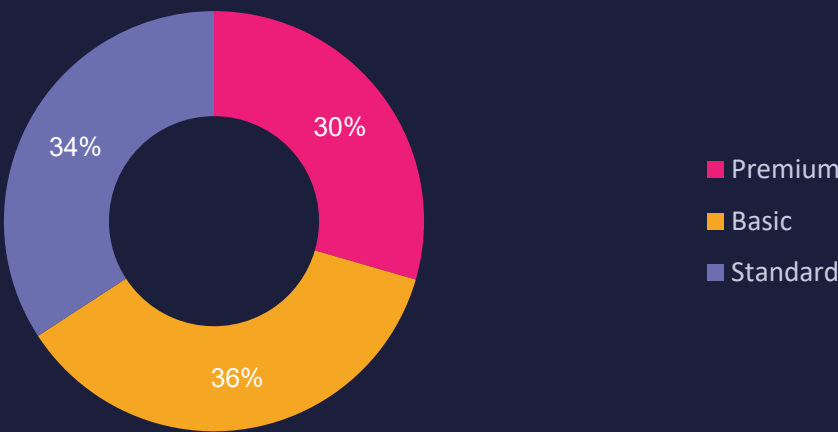
Where NovaTel stands today, at a glance



Churn Rate by Contract Type

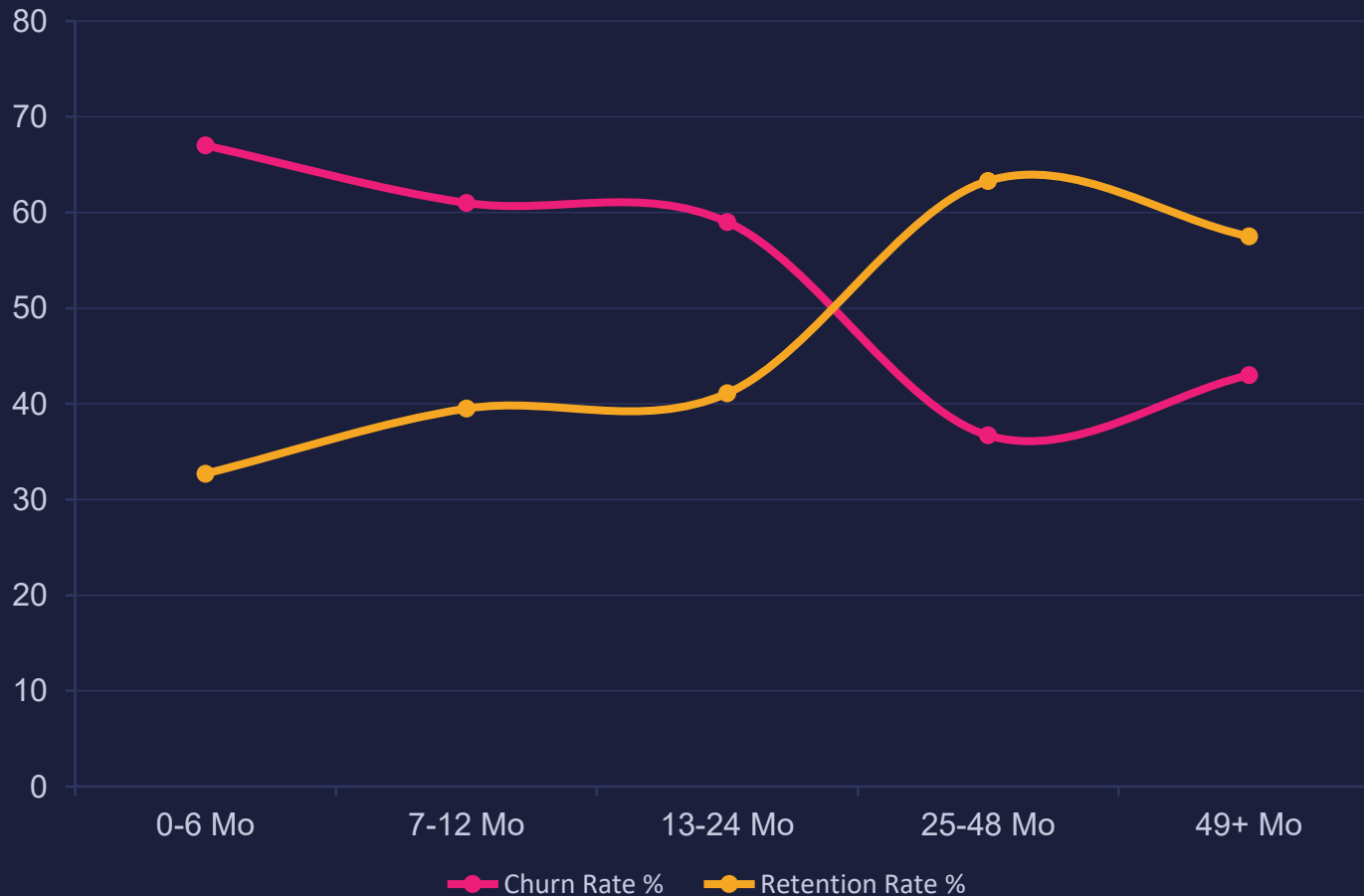


Churned Customers by Plan Type



Insight: The Retention Opportunity Sits Early

Retention improves sharply with tenure — the first 6 months are the danger zone



What this tells us

- 25-48 month customers retain at 63.3% — the highest of any tenure band, and 93.7% better than 0-6 month customers.
- 0-6 month customers retain at only 32.7%, the lowest across all bands — the earliest relationship stage is the weakest.
- This is a solvable, structural problem: better onboarding and early engagement can move customers past the highest-risk window faster.

Insight: Tracing Churn Back to Its Root Cause

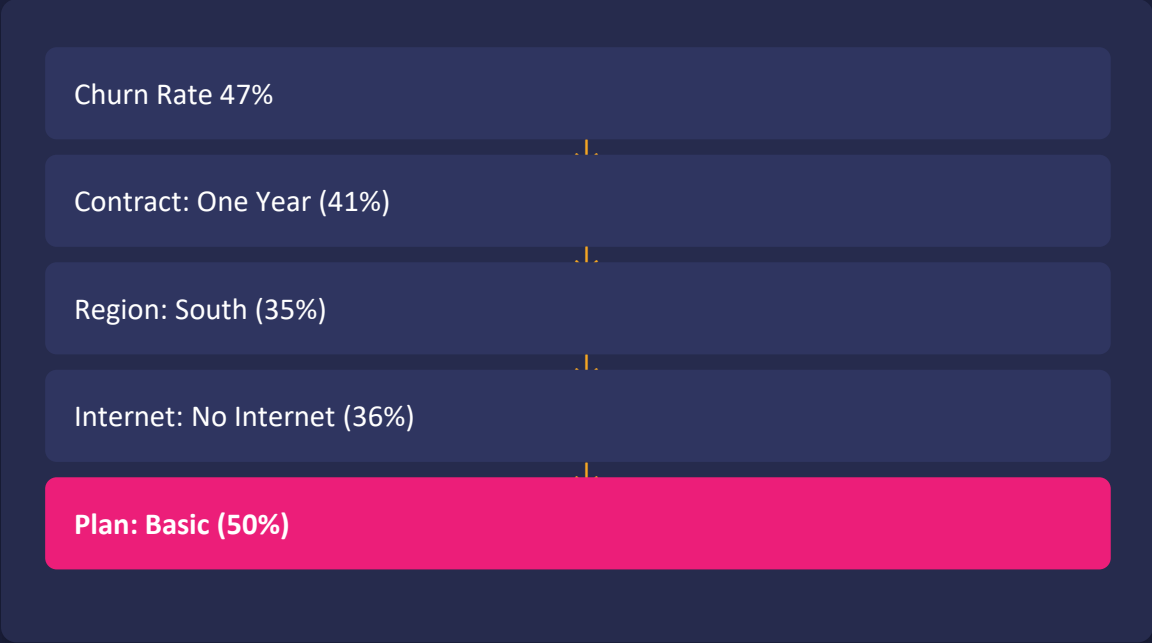
Two diagnostic views that let a stakeholder ask their own follow-up question

Churn Risk Funnel



46.8% of all customers who raise a complaint or place a support call go on to churn — a clear intervention point before the loss happens.

Decomposition Tree: Drill Any Path

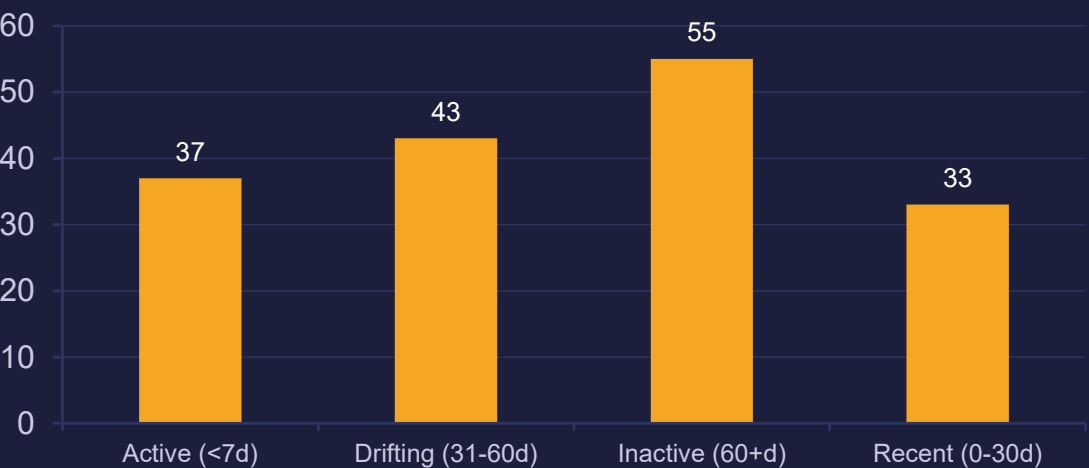


A stakeholder can click through Contract → Region → Internet Service → Plan Type in any order to isolate the exact customer segment driving a spike — no separate report request needed.

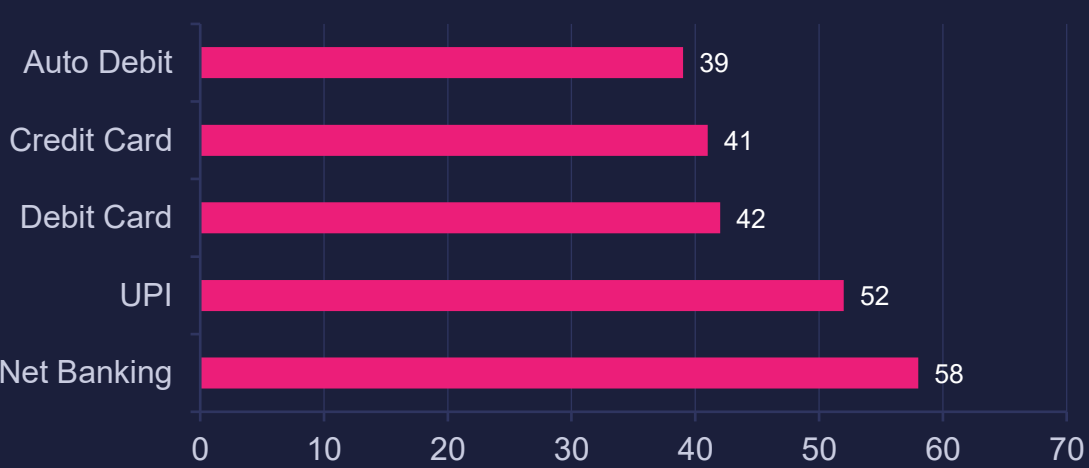
Insight: Churn Gives Warning Before It Happens

Login inactivity and digital payment methods are the clearest early signals

Churn Rate by Login Activity



Churn Rate by Payment Method



Actionable pattern: Customers inactive for 60+ days churn at 55% — 68.7% higher than recently active customers. Net Banking and UPI users churn well above card users, suggesting payment friction or failed-transaction experiences may be an overlooked driver worth investigating directly.

Built for Self-Service, Not a Static Snapshot

Stakeholders can reshape the analysis themselves, without waiting on a rebuild



What-If Target Sliders

Target Churn Rate % (13%) and Target Retention Rate % (85%) are live parameters — drag either slider and every KPI, gauge, and chart recalculates against the new goal instantly.



Dynamic Metric Selector

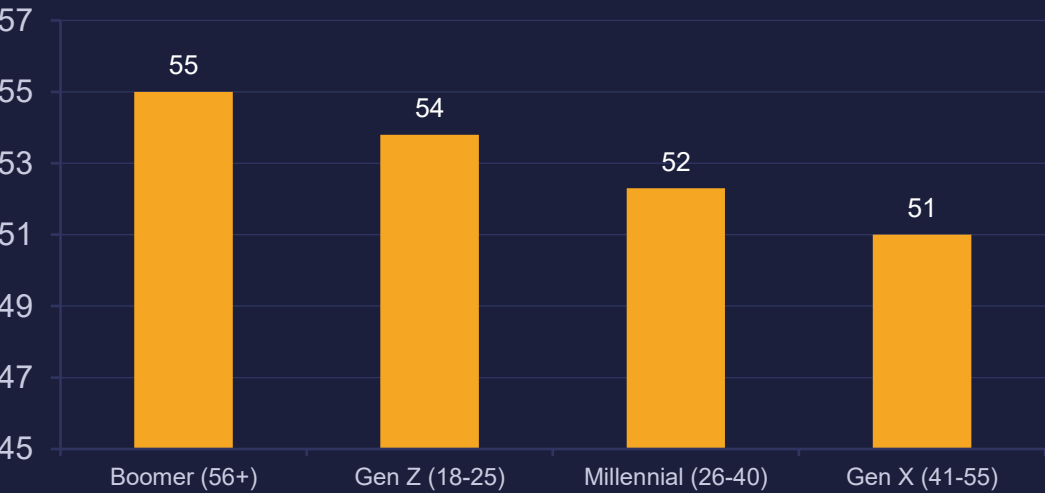
A single chart toggles between Churn Rate % and Retention Rate % (or shows both) via a Filter Parameter — one visual instead of two redundant ones.



Global Filter Panel

Region, Contract_Type, Plan_Type, and Last_Login_Days slicers let any stakeholder narrow entire report to their own scope in one click.

Retention Rate % by Age Group



Demographic read: Retention is fairly even across age groups (51-55%) — age is not a major churn driver here, which helps rule it out and keep retention spend focused on contract type, tenure, and engagement instead.

Priority Segment: High-Value Customers at Risk

Where retention spend will have the greatest financial impact



147

At-Risk High-Value Customers



22.3K

Avg. Total Charges Retained (per customer)



102.24K

Total Revenue at Risk

Sample: Highest-Priority Accounts for Outreach

Customer ID	Plan	Monthly Charges	Support Calls	Last Login (days)
NVT00309	Basic	\$321.01	10	20
NVT00190	Basic	\$344.24	10	98
NVT00240	Basic	\$354.23	10	30
NVT00006	Premium	\$810.74	10	29
NVT00207	Premium	\$818.04	10	89

Recommendations

Prioritized actions the retention team can take this quarter



Target the 147 high-value at-risk accounts first

Highest revenue protected per outreach dollar spent — start here for fastest ROI.



Launch a 0-6 month onboarding program

Retention in this band is the weakest (32.7%) — closing this gap has the largest structural upside.



Incentivize contract migration

Move Month-to-Month customers (68% churn) toward One/Two-Year plans (35-41% churn) with loyalty pricing.



Trigger win-back outreach at 60 days inactive

Login inactivity is the strongest early warning signal available — automate proactive contact before churn happens.



Investigate Net Banking / UPI payment friction

These payment methods show the highest churn — worth a direct root-cause review with the payments team.

Thank You

Questions & Discussion

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